



AI SECURITY SHIELD

CYBER DEFENDERS

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PROBLEM STATEMENT:



Threat Intelligence Correlation & Alert Prioritisation Assistant

Users encounter suspicious emails, malicious links, authentication anomalies, and endpoint compromises every day. Existing security tools generate fragmented, siloed alerts that require security analysts to manually correlate them - by which time an attack chain may already be in progress.



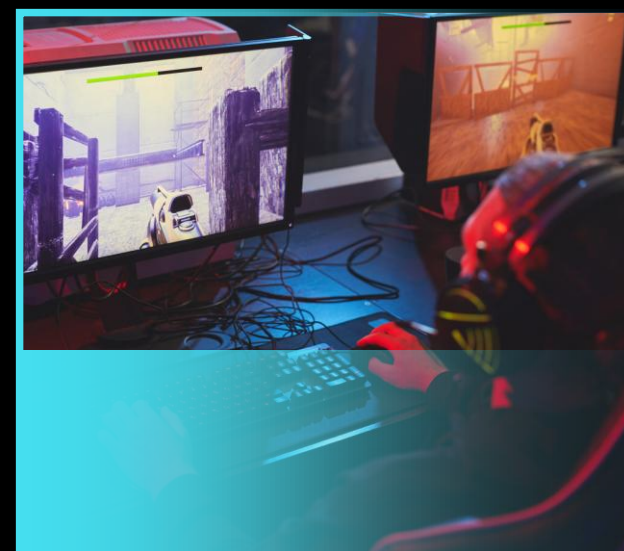


THE INDUSTRY PROBLEM:

THE PROBLEM:

Alerts are fragmented across different systems
Multiple alerts may actually belong to one attack
Analysts spend time manually connecting events
False positives consume investigation time
Critical threats can get buried in alert volume

“THE CHALLENGE ISN'T GENERATING MORE ALERTS. IT'S IDENTIFYING WHICH ALERTS MATTER.”





OUR SOLUTION:

AI Security Shield ingests simulated security signals, runs them through a multi-stage analysis pipeline, and produces correlated incidents with AI-powered explanations:

Security Signals (Email → URL → Auth
→ Endpoint → Threat Intel)

- * Normalization

- * Correlation (shared user/device/IP/ domain + time proximity)

- £ Risk Scoring (5-factor transparent formula, 0-100)

- * Technique Classification (MITRE ATT&CK references)

- \ AI Investigation Summary (IBM watsonx.ai or built-in fallback)

- V Security Shield Warning (HIGH/ CRITICAL incidents)





AI Security Shield acts as an intelligent warning layer. It collects security signals, understands their context, identifies suspicious patterns across signals, assigns a transparent risk score, explains why something is risky, and fires an immediate, prominent Security Shield Warning before threats escalate.

AI Security Shield transforms isolated security alerts into prioritized, explainable threat incidents.





HOW IT WORKS:



1. INGEST

Bring security signals together*

- * Email alerts & URL indicators
- * Authentication events
- * Endpoint activity & Threat intelligence

2. CORRELATE

Connect the dots*

AI identifies relationships using:

- * Common entities & Time proximity
- * Indicators & patterns
- * User/device behaviour & Threat intelligence

3. PRIORITIZED

Find what matters first

AI evaluates:

Risk + Confidence + Severity

LOW → MEDIUM → HIGH → CRITICAL*

1.-EXPLAIN

Turn signals into actionable intelligence*

- * MITRE ATT&CK mapping
- * Supporting evidence
- * BLUF investigation summary
- * Recommended next steps





THE INNOVATION



We don't just detect alerts. We connect the dots.

Multiple Alerts



☐ *AI Correlation*



☐ *Connected Incident*



☐ *Critical Threat*

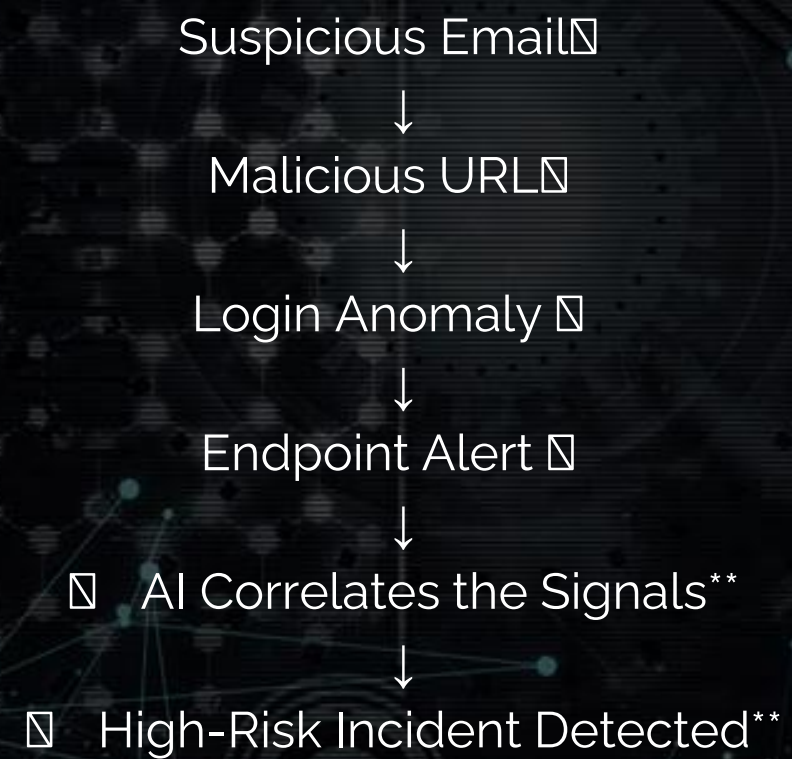
Traditional: 500 alerts → Analyst checks individually

AI Security Shield: 500 alerts → Correlates → Prioritizes → Focus on what matters first





DEMO SCENARIO



MITRE ATT&CK → BLUF → Recommended Action



EXPLAINABLE THREAT INTELLIGENCE



Know why a threat matters.

▣ Risk Score

92/100 — CRITICAL

▣ Evidence

Correlated alerts + threat indicators

▣ MITRE ATT&CK

Mapped attack technique

▣ BLUF

Clear incident summary + next steps

From detection → to understanding → to action.





SECURITY SHIELD

Protect users before they become victims.

☒ Suspicious Interaction Detected



☒ AI Assesses the Risk



⚠☒ Instant Security Warning



☒ ☒ User Takes Action

No manual copy-paste. No waiting for investigation.

AI Security Shield warns at the point of risk.





TECHNOLOGY & IBM BOB

Built for speed. Designed for security.

BUILT FOR SPEED. DESIGNED FOR SECURITY.

- ❑ IBM Bob — AI-powered development
- ❑ React / Vite — Frontend
- ❑ Python / FastAPI — Backend
- ❑ Simulated Security Data — Alerts & Threat Intel
- ❑ MITRE ATT&CK — Threat Mapping

Simple Architecture → Fast MVP → Explainable Security



IMPACT & FUTURE

From Alert Overload → Actionable Intelligence

- ❑ Reduce alert fatigue
- ❑ Prioritize critical threats
- ❑ Explain why threats matter
- ❑ Enable faster response

Future:

- ❑ Real SIEM & EDR Integration
- ❑ Live Threat Intelligence
- ❑ Advanced AI Threat Detection
- ❑ Proactive Enterprise Protection

AI Security Shield — Connect. Prioritize. Protect.





THANK YOU!

- CYBER DEFENDERS

